

6.6 Summary

Here's a crisp **summary** of the tools and technologies used in the **SHIFTY project**:

1. **React Native** is used for cross-platform mobile app development, ensuring a seamless experience on both iOS and Android.
2. **Redux Toolkit** manages the app's state efficiently, while **Axios** handles API requests.
3. The backend is powered by **Node.js with Express.js**, offering scalability and fast request handling.
4. **Python** is used for machine learning models, with **Scikit-learn** (Random Forest) and **XGBoost** for dynamic pricing.
5. ML models are deployed as REST APIs using **FastAPI/Flask**.
6. **MongoDB** serves as the primary database, handling flexible inventory and booking data.
7. **Rasa** with **SpaCy** powers the AI chatbot for conversational support.
8. Cloud deployment is done on **AWS/GCP**, using **Docker** for containerization and **Kubernetes** for orchestration.
9. **GitHub Actions/Jenkins** provides a CI/CD pipeline for automated builds and deployments.
10. **Postman, VS Code, and PyCharm** are used for development and testing.
11. Security is ensured with **JWT, OAuth2.0, AES-256 encryption, and TLS certificates** for safe transactions.